

SmartLoad Project – Progress Update Report

1. Meeting Overview

we wanted to share a summary of what was discussed about our project **SmartLoad**. The meeting focused mainly on the challenges related to the dataset, defining the project scope, and listing the main requirements.

During the discussion, we also drew an initial sketch of the proposed system architecture. It shows the general workflow of the system: starting from how the traffic enters the load balancer and passes through the API gateway to the microservices layer, then how telemetry data is collected and used by the AI module for autoscaling and load optimization.

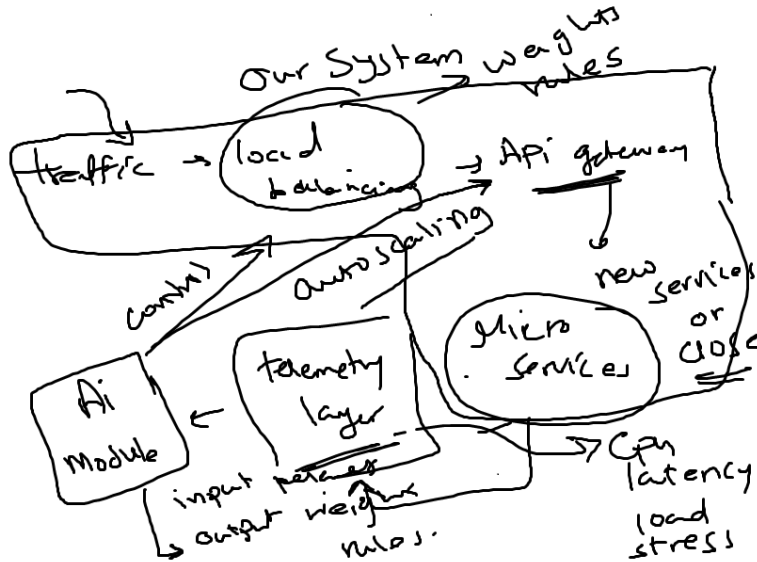


Figure 1: Preliminary system architecture sketch of SmartLoad.

2. Supervisor Feedback and Key Requirements

Dr. Doaa advised us to focus on the **high-level design** rather than diving too deeply into each module at this stage. She suggested that we first agree on the overall structure and layers of the system, and later, we can go deeper into each part (like algorithms, technologies, or data models).

Dr. Doaa outlined the following main tasks for the coming period:

1. **High-level System Design:** Create a clear architecture diagram and describe how the main components interact.
2. **Main Features List:** Define the core functions SmartLoad will provide, such as adaptive load balancing, predictive scaling, and anomaly detection.
3. **Market Survey:** Review existing tools (NGINX, AWS Load Balancer, Kubernetes) to understand what they offer and their limitations.
4. **Comparison with the Market:** Highlight how SmartLoad stands out in terms of simplicity, cost, and AI integration.
5. **Address the Gap:** Explain the unique value SmartLoad adds to both industry and research.

3. Project Direction

Dr. Doaa also encouraged us to start by focusing on the **industry aspect** of SmartLoad — treating it as a practical product or middleware that small and medium companies could use.

Once we have a solid and realistic system structure, we can extend it toward the **research side**, where our contribution could be the integration of all layers into one intelligent system or proposing a new algorithm that supports adaptive workload balancing.

She emphasized that this stage should be more about the concept and big picture — how the system works and what makes it useful — rather than about implementation or specific AI models.

4. Next Steps

Task	Description
High-Level System Design	Refine the architecture diagram and explain each major layer in simple terms.
Main Features List	Summarize SmartLoad’s main features and use cases.
Market Survey	Collect information about existing solutions and their limitations.
Comparison & Gap Analysis	Show how SmartLoad can fill the current gaps in the market.